

Before changes were made:

```

javier-debian% grep "Port" /etc/ssh/sshd_config
#Port 22
#GatewayPorts no
javier-debian% grep "PermitRootLogin" /etc/ssh/sshd_config
#PermitRootLogin prohibit-password
# the setting of "PermitRootLogin prohibit-password".
javier-debian% cat /etc/motd

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
javier-debian% ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s31f6: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 22:3c:42:ad:bb:20 brd ff:ff:ff:ff:ff:ff permaddr 9c:6b:00:6e:0c:f6
    altname enx9c6b006e0cf6
    inet 192.168.4.48/22 brd 192.168.7.255 scope global dynamic noprefixroute enp0s31f6
        valid_lft 14180sec preferred_lft 14180sec
    inet6 fd0b:43da:5d03:1:5bee:d07b:96d7:3be3/64 scope global temporary dynamic
        valid_lft 604583sec preferred_lft 86013sec
    inet6 fd0b:43da:5d03:1:203c:42ff:fead:bb20/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 2591997sec preferred_lft 604797sec
    inet6 fe80::203c:42ff:fead:bb20/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: wlp4s0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default qlen 1000
    link/ether 76:72:d3:b3:2c:65 brd ff:ff:ff:ff:ff:ff permaddr 78:46:5c:91:9a:49
    altname wlx78465c919a49
javier-debian% 
```

```

GNU nano 8.4 /etc/ssh/sshd_config
# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/bin:/usr/bin:/bin:/usr/games

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin prohibit-password
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

#PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
#AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
```

```
Apr 29 1:47 PM
Terminal
/etc/ssh/sshd_config *

GNU nano 8.4
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to "no" here!
#PasswordAuthentication yes
#PermitEmptyPasswords no

# Change to "yes" to enable keyboard-interactive authentication. Depending on
# the system's configuration, this may involve passwords, challenge-response,
# one-time passwords or some combination of these and other methods.
# Beware issues with some PAM modules and threads.
KbdInteractiveAuthentication no

# Kerberos options
#KerberosAuthentication no
#KerberosDRLocalPasswd yes
#KerberosTicketCleanup yes
#KerberosGetAFSToken no

# GSSAPI options
#GSSAPIAuthentication no
#GSSAPICleanupCredentials yes
#GSSAPIStrictAcceptorCheck yes
#GSSAPIKeyExchange no

# Set this to 'yes' to enable PAM authentication, account processing,
# and session processing. If this is enabled, PAM authentication will
# be allowed through the KbdInteractiveAuthentication and
# PasswordAuthentication. Depending on your PAM configuration,
# PAM authentication via KbdInteractiveAuthentication may bypass
# the setting of "PermitRootLogin prohibit-password".
# If you just want the PAM account and session checks to run without
# PAM authentication, then enable this but set PasswordAuthentication
# and KbdInteractiveAuthentication to 'no'.
UsePAM yes

#AllowAgentForwarding yes
#AllowTcpForwarding yes
#GatewayPorts no
X11Forwarding yes
#X11DisplayOffset 10
#X11UseLocalhost yes
#PermitTTY yes
PrintMotd no
#PrintLastLog yes
#TCPKeepAlive yes
#PermitUserEnvironment no
#Compression delayed
#ClientAliveInterval 0
#ClientAliveCountMax 3
```

```
#UseDNS no
#PidFile /run/sshd.pid
#MaxStartups 10:30:100
#PermitTunnel no
#ChrootDirectory none
#versionAddendum none

# no default banner path
#Banner none

# Allow client to pass locale and color environment variables
AcceptEnv LANG LC_* COLORTERM NO_COLOR

# override default of no subsystems
Subsystem sftp /usr/lib/openssh/sftp-server

# Example of overriding settings on a per-user basis
Match User anoncvs
# X11Forwarding no
# AllowTcpForwarding no
# PermitTTY no
# ForceCommand cvs server
```

Changes that were made

Updated Banner:

```
Apr 29 1:44 PM
Terminal
/etc/ssh/banner

GNU nano 8.4
*****
*                Welcome to the SSH Lab VM!                *
*                Only authorized lab users may connect.      *
* Lab for demonstration purposes in Professor Sullwold's class *
*****
```

Updated MOTD:

```

Apr 29 1:40 PM
Terminal
GNU nano 8.4 /etc/motd *

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My name is Javier Silva, and my ID number is 0587949
I just completed Part 1 of the lab.
```

Updated SSH Config:

```
Port 1622
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
PermitRootLogin yes
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

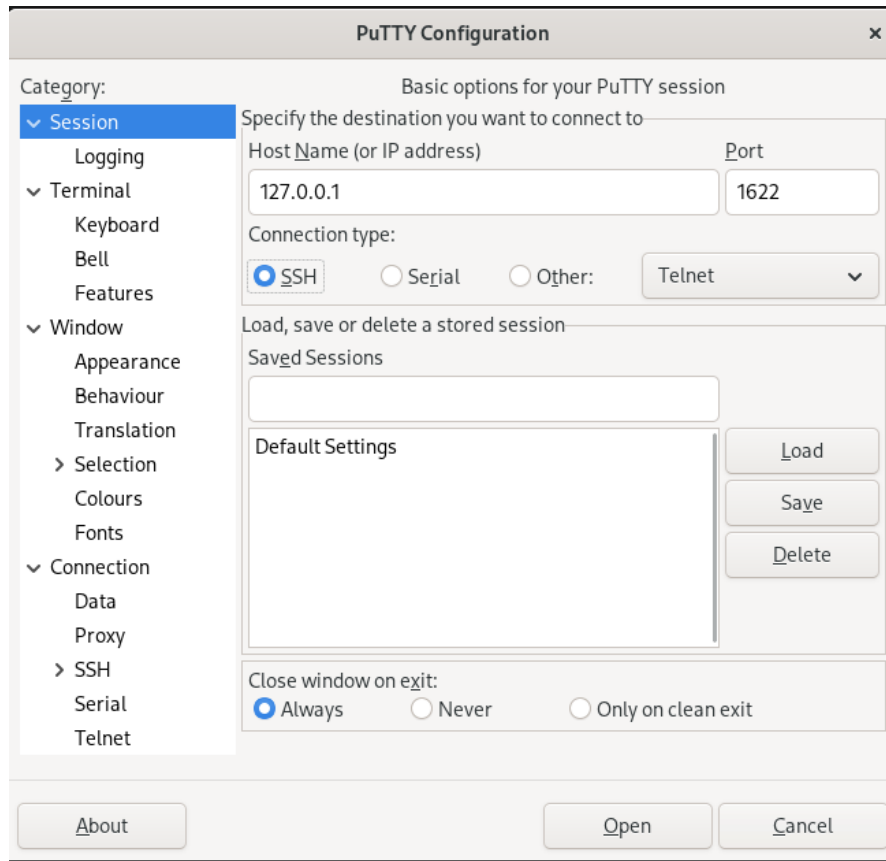
# To disable tunneled clear text passwords, change to "no" here!
PasswordAuthentication yes
#PermitEmptyPasswords no

# no default banner path
Banner /etc/ssh/banner
```

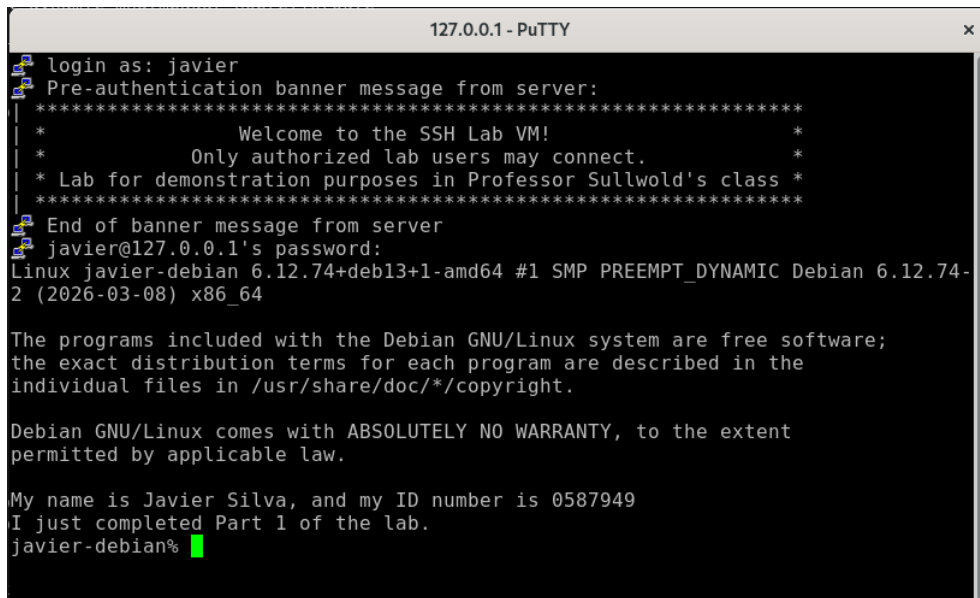
```
javier-debian% sudo systemctl restart ssh
javier-debian% ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s31f6: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 22:3c:42:ad:bb:20 brd ff:ff:ff:ff:ff:ff permaddr 9c:0b:00:0e:0c:f6
    altname enx9c0b00e0cfc6
    inet 192.168.4.48/22 brd 192.168.7.255 scope global dynamic noprefixroute enp0s31f6
        valid_lft 12334sec preferred_lft 12334sec
    inet6 100b:430a:5003:1:203c:42ff:fead:bb20/64 scope global temporary dynamic
        valid_lft 602737sec preferred_lft 84167sec
    inet6 fd0b:430a:5003:1:203c:42ff:fead:bb20/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 2591642sec preferred_lft 604442sec
    inet6 fe80:430a:5003:1:203c:42ff:fead:bb20/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: wlp4s0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default qlen 1000
    link/ether 9a:f9:71:d3:e9:f3 brd ff:ff:ff:ff:ff:ff permaddr 78:46:5c:91:9a:49
    altname wlx78465c919a49
javier-debian% █
```

PART 1

```
# To disable tunneled clear text passwords, change to "no" here!  
PasswordAuthentication yes  
PermitEmptyPasswords yes
```



The image shows the PuTTY Configuration dialog box. The 'Session' category is selected in the left sidebar. The main area is titled 'Basic options for your PuTTY session'. It contains fields for 'Host Name (or IP address)' set to '127.0.0.1' and 'Port' set to '1622'. The 'Connection type' is set to 'SSH' with radio buttons for 'SSH', 'Serial', and 'Other:'. Below this is a section for 'Load, save or delete a stored session' with a list of 'Saved Sessions' containing 'Default Settings' and buttons for 'Load', 'Save', and 'Delete'. At the bottom, there are options for 'Close window on exit' with radio buttons for 'Always', 'Never', and 'Only on clean exit'. The 'About', 'Open', and 'Cancel' buttons are at the bottom of the dialog.

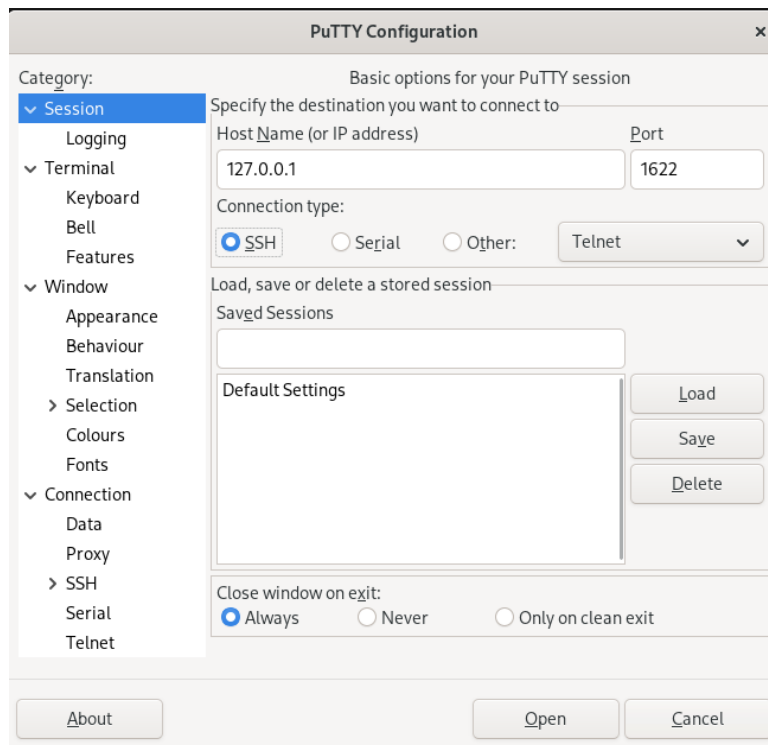


The image shows a terminal window titled '127.0.0.1 - PuTTY'. It displays the output of an SSH connection. The prompt is 'login as: javier'. A pre-authentication banner message from the server is shown, followed by the user's password prompt. The terminal output shows the user is logged in as 'javier' on a Debian system. The banner message reads: 'Welcome to the SSH Lab VM! Only authorized lab users may connect. Lab for demonstration purposes in Professor Sullwold's class'. The terminal also shows the Debian GNU/Linux version and the user's name and ID number.

```
login as: javier  
Pre-authentication banner message from server:  
*****  
*           Welcome to the SSH Lab VM!           *  
*   Only authorized lab users may connect.   *  
* Lab for demonstration purposes in Professor Sullwold's class *  
*****  
End of banner message from server  
javier@127.0.0.1's password:  
Linux javier-debian 6.12.74+deb13+1-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.74-  
2 (2026-03-08) x86_64  
  
The programs included with the Debian GNU/Linux system are free software;  
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individual files in /usr/share/doc/*/copyright.  
  
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permitted by applicable law.  
  
My name is Javier Silva, and my ID number is 0587949  
I just completed Part 1 of the lab.  
javier-debian% 
```

Part 2

```
# To disable tunneled clear text passwords, change to "no" here!  
PasswordAuthentication yes  
PermitEmptyPasswords yes
```



```
127.0.0.1 - PuTTY  
login as: sshTest  
Pre-authentication banner message from server:  
*****  
*           Welcome to the SSH Lab VM!           *  
*   Only authorized lab users may connect.   *  
* Lab for demonstration purposes in Professor Sullwold's class *  
*****  
End of banner message from server  
Linux javier-debian 6.12.74+deb13+1-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.74-  
2 (2026-03-08) x86_64  
  
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permitted by applicable law.  
  
My name is Javier Silva, and my ID number is 0587949  
I just completed Part 2 of the lab.  
$
```

Part 3

```
# To disable tunneled clear text passwords, change to "no" here!  
PasswordAuthentication no  
PermitEmptyPasswords no
```

PuTTY Configuration

Category: Basic options for your PuTTY session

Session: Specify the destination you want to connect to

Host Name (or IP address): 127.0.0.1 Port: 1622

Connection type: ☒ SSH ☐ Serial ☐ Other: Telnet

Window: Load, save or delete a stored session

Appearance: Saved Sessions

Behaviour: Default Settings

Translation: Load Save Delete

Selection: Close window on exit: ☒ Always ☐ Never ☐ Only on clean exit

Colours: About Open Cancel

Fonts: Connection

Data: Public-key authentication

Proxy: Private key file for authentication: /home/javier/lab_key.ppk Browse...

SSH: Certificate to use with the private key (optional): Browse...

Kex: Plugin to provide authentication responses

Host keys: Plugin command to run

Cipher: Credentials

Auth: GSSAPI

TTY

X11

Tunnels

About Open Cancel

```
login as: javier
Pre-authentication banner message from server:
*****
*               Welcome to the SSH Lab VM!               *
*       Only authorized lab users may connect.           *
* Lab for demonstration purposes in Professor Sullwold's class *
*****
End of banner message from server
Authenticating with public key "rsa-key-20260429"
Linux javier-debian 6.12.74+deb13+1-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.74-
2 (2026-03-08) x86_64

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My name is Javier Silva, and my ID number is 0587949
I just completed Part 3 of the lab.
Last login: Wed Apr 29 14:38:53 2026 from 127.0.0.1
javier-debian% 
```

COMMAND EXPLANATIONS

Part 1

sudo nano /etc/ssh/banner

- **Explanation:** Creates the custom text file to serve as the pre-login warning banner. Necessary to fulfill the lab requirement for a pre-login warning.

sudo nano /etc/motd

- **Explanation:** Modifies the "Message of the Day" file. Necessary to display the required student name, ID, and Part 1 completion status after a successful login.

sudo nano /etc/ssh/sshd_config

- **Explanation:** Opens the primary SSH daemon configuration file. Necessary to change the listening port to 1622, enable root login, enforce password authentication, and point the service to the custom banner file.

sudo systemctl restart ssh

- **Explanation:** Restarts the SSH service. Necessary so the configuration changes made in sshd_config take effect.

ip addr

- **Explanation:** Displays network interface information. Necessary to identify the IP address of the Linux VM to connect to it via PuTTY from the host machine.

Part 2

sudo useradd -m sshTest

- **Explanation:** Creates a new user account named sshTest and generates a home directory (-m) for them. Necessary to fulfill the requirement of creating the specific lab user.

sudo passwd -d sshTest

- **Explanation:** Deletes the password for the sshTest account (-d flag), leaving it blank. Necessary to allow authentication without a password.

sudo nano /etc/ssh/sshd_config (Part 2 update)

- **Explanation:** Modified the configuration to set PermitEmptyPasswords yes. This is necessary because SSH blocks accounts with blank passwords by default; this explicitly overrides that security measure.

Part 3

mkdir -p ~/.ssh

- **Explanation:** Creates the .ssh directory in the user's home folder if it doesn't already exist. Necessary to hold the authorized keys file.

nano ~/.ssh/authorized_keys

- **Explanation:** Creates/edits the file that stores approved public SSH keys. Necessary to tell the SSH server to trust the specific key generated by the client machine.

chmod 600 ~/.ssh/authorized_keys

- **Explanation:** Restricts read and write permissions to the owner only. Necessary because the SSH service will reject key files that are accessible by other users for security reasons.

sudo nano /etc/ssh/sshd_config (Part 3 update)

- **Explanation:** Modified the configuration to set PasswordAuthentication no and PermitEmptyPasswords no. Necessary to enforce key-based authentication exclusively and meet the requirement of demonstrating that passwords no longer work.